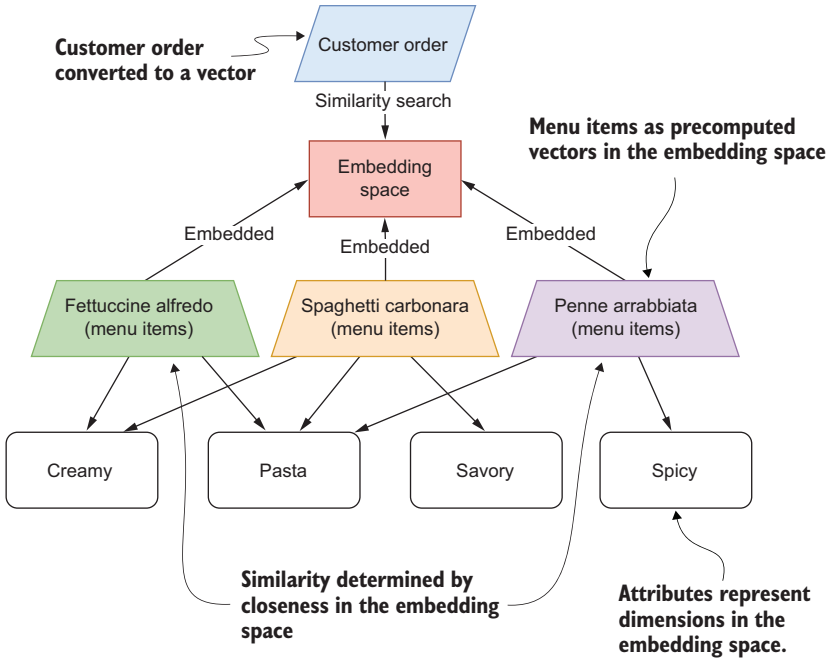


Figure 5.8 The concept of embeddings using a restaurant menu analogy. The "embedding space" represents a multidimensional space where both customer orders (queries) and menu items are mapped as vectors. Each menu item (e.g., spaghetti carbonara) is a precomputed vector, while the customer order is converted to a query vector. Similarity between items is determined by their proximity in this space. Attributes like "pasta" or "creamy" represent dimensions of the embedding space. A similarity search finds the closest match and items similar to the query, mirroring how embedding-based systems in machine learning find relevant results for given inputs.

Here's a breakdown of figure 5.8:

- *Customer order as query vector*—The "customer order" is represented as a query vector in this space. This is analogous to how a user's input or search query would be converted into a vector in an embedding-based system.
- *Menu items as vectors*—Each menu item (spaghetti carbonara, fettuccine alfredo, and penne arrabbiata) is also represented as a vector in the same space. These would be precomputed embeddings of known items in the system.
- *Similarity search*—A similarity search is performed by comparing the customer order vector to the menu item vectors within the embedding space. This represents how embedding-based systems find similar items or relevant responses.
- *Closest match and similar items*—The system can rank results based on their similarity to the query by comparing their distances.
- *Attributes*—Each menu item is connected to various attributes ("pasta," "creamy," "savory," "spicy"). These attributes can be thought of as dimensions or features in the embedding space that contribute to the position of each item's vector.